How Much Dependency Can a Marginal Rule See? Total Correlation in Diffusion LM Parallel Decoding

Dovud Asadov Rustambek Urokov Qobiljon Toshnazarov
Kayumov Abduaziz Mardon Hazratov Husanboy Mansuraliyev
New Uzbekistan University
{d.asadov, r.urokov, q.toshnazarov, a.kayumov, m.hazratov, h.mansuraliyev}@newuu.uz

Abstract

A masked diffusion language model commits several masked positions per step. When its conditionals compose into a joint, the exact cost of committing a set S is that joint's conditional total correlation $\mathrm{C}(S \mid c)$. When they do not, we say exactly what our estimate measures. Many systems choose the commit set from per-position marginals. That such a rule cannot *determine* C is elementary, and we state it exactly: the marginals certify $\mathrm{C} \le W = B - \max_i H(Y_i)$, with B the summed marginal entropy, and at fixed marginals C takes every value up to a maximum $\mathrm{C}_{\max} \le W$. We ask instead how well the marginals *predict* an order-averaged estimate $\widehat{\mathrm{C}}$ of C, per candidate set on four masked diffusion LMs from two families, 247 to 516 blocks per model and regime. Outside Dream-instruct, a marginal-only predictor scored out-of-fold and corrected for estimator variance leaves a residual of 0.74 to 1.17 bits, 1.8 to 3.7 times below a reference that knows the marginal entropies but not the coupling. Most of that predictability is the window itself. On a subset, $\widehat{\mathrm{C}}$ ranks blocks like the KL cost of a parallel commit but sits a median 0.4 to 1.1 bits below it on prose and in the deployed regime. Dream-instruct is the exception, mainly on prose, where its conditionals are least self-consistent and the estimate stops measuring a commit cost. These results concern predicting C for a fixed set, not the gain of a decoder that coordinates its commits. Getting there requires a validated readout. Dream and LLaDA need *opposite* logit-position conventions, and reading Dream with the wrong one costs 6.9 bits on the correct token, inflates a conditional-inconsistency diagnostic 14 to 41×, and in the deployed regime erases the confidence-error correlation on Dream-base and reverses it on Dream-instruct. We specify gates that measure the convention and report them for every checkpoint. Scoring six marginal rules against measured C, pooling across the block beats gating on its weakest position in all 24 paired comparisons, and the top-2 margin, which dominates confidence on tasks with analytic C, does not on these checkpoints.

1 Introduction

A masked diffusion language model (dLLM) [1, 2] writes text by committing several masked positions per denoising step, and that parallelism is the entire source of its speed. Which positions to commit is therefore the central control decision, and a large body of work answers it with a rule computed from each position's marginal separately: token confidence [4, 5], the top-2 margin, the predictive entropy, or cross-step stability [6, 19]. A second line estimates the dependency itself, usually from attention [7, 8].

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Diffusion Language Models: Foundations, Efficiency, and Reasoning (DiffuLM).

The cost these rules are approximating is known exactly. Committing a set S in one step samples each position from its own marginal and discards the joint structure, and the Kullback–Leibler cost of that factorization is the *conditional total correlation*

$$\mathrm{C}(S \mid c) \;=\; \sum_{i \in S} H(Y_i \mid c) \;-\; H(Y_S \mid c) \;\geq\; 0, \tag{1}$$

the information shared among the positions of S given the already-decoded context c [9, 10]. It is zero exactly when the positions are conditionally independent, and it lower-bounds the achievable KL of a parallel commit [9, 17].

That a marginal rule cannot *determine* C is not news: it is stated in [20], motivates several decoders [18, 22, 28], and in the worst case takes two lines of information theory, which §3 states exactly as a reference point. But the worst case is not the question a practitioner has, since a decoder only meets the joints a trained model produces. The open question, and the one we answer, is:

> *On a real dLLM, how much of the variation in* C *is recoverable from the marginals, and how much do the rules that read them leave behind?*

Answering it needs two things the literature has not put together: C measured per candidate set on a real checkpoint, and a readout of that checkpoint one can trust.

Reading a dLLM's marginals is not trivial, and getting it wrong is not loud. Dream [2] keeps the shift of the autoregressive backbone it was initialised from, so its hidden state at i predicts the token at $i{+}1$. LLaDA [1] predicts masked tokens in place. Two of the most widely used open checkpoints require *opposite* conventions, and the difference is invisible to any type check. Reading Dream at $\delta{=}0$ costs 6.9 bits on the correct token. Its greedy decodes then collapse into a repeated token, but a measurement pipeline that never decodes text sees none of that, and the confidence-error correlation it reports is erased or reversed without looking wrong (§5). We therefore specify three gates that *measure* the convention rather than assuming it.

Contributions. **(1)** A measurement protocol with three gates (§4), namely readout offset, forward-pass determinism, and a decode-and-read check. The offset gate scores *hidden* true tokens, so a bidirectional model cannot pass by copying, and it recovers each checkpoint's documented convention without being told it. **(2)** A single-variable ablation of what a wrong readout does to the measurement (§5). **(3)** C estimated per candidate set on four checkpoints, with the residual left by the strongest marginal-only predictor we could fit, scored out-of-fold and corrected for estimator variance, and a precise account of what the estimate means when the conditionals do not compose (§6). **(4)** All six deployed rules against measured C, with paired tests (§7). As a reference point we state the worst case exactly (§3), a short characterisation built on the factorisation bound of [17, Thm. 1] and the analysis of [9, §4].

2 Background and related work

Masked diffusion decoding. A masked dLLM's one-step kernel over the masked positions factorises, $p_\theta(x_{t-1} \mid x_t) = \prod_i p_\theta(x^i_{t-1} \mid x_t)$, a mean-field product [12, 13]. A decoder reveals a subset S per step, sampling each position from its marginal, and $|S| = 1$ recovers any-order autoregression.

Total correlation as the cost of parallelism. That C governs parallel-commit error is established from several directions: a KL floor [9, 17], the dependency a factorised step drops [10], sequence-error step-scaling [14], optimal-schedule analysis [16], convergence rates in C and its dual [26], an information-theoretic separation of generation order from parallelisation bias [11], and a Gaussian-field analysis [23]. Dataset-level estimation dates to non-autoregressive translation [17]. ParallelBench [9] supplies the synthetic family with analytic C that we reuse in §3, and finds that a confidence rule loses to random selection there once $\mathrm{C} > 0$. We add a per-candidate-set estimate of C on real checkpoints and ask how much of it the marginals recover.

Dependency-aware decoding. Many recent decoders replace confidence with an estimated dependency signal: attention-derived graphs [7, 8, 29], learned pairwise mutual information [21], dependency predictors [22, 27, 28], mean-field coordination [30], conditional independence testing

[36], local coherence [37], a learned commitment policy [31], and a coordination pass that names C as the obstacle [20]. We measure how well the marginals predict the dependency of a fixed candidate set, not how much a decoder that coordinates its commits would gain, which needs a matched-compute comparison that we do not run.

Conditional inconsistency in masked models. That an MLM's conditionals need not compose into any joint is long established [33], with consequences for sampling dynamics [34] and denoising order [35]. The specific diagnostic we use as a covariate, which holds the assignment fixed and varies the reveal order so that a consistent joint gives exactly zero spread, is *chain-rule deviation*, introduced for order-agnostic LMs by [32]. We do not claim it, and report it as a covariate that §5 shows is highly sensitive to a readout error.

3 The worst case: what a marginal profile certifies

Since $H(Y_S \mid c) \geq 0$, Eq. (1) gives immediately

$$0 \;\leq\; \mathrm{C}(S \mid c) \;\leq\; \sum_{i \in S} H(Y_i \mid c) \;=:\; B(S \mid c), \tag{2}$$

so the summed marginal entropy is a *certified upper bound* on the dependency a parallel commit discards. A tighter ceiling that depends on the marginals only through their entropies is $\mathrm{C} \leq W := B - \max_i H(Y_i \mid c)$, since a joint is never less uncertain than its most uncertain coordinate, and it shows exactly when the marginals settle the question: they certify $\mathrm{C} = 0$ if and only if at most one position of S is uncertain.

The set of joints with a fixed marginal profile (a transportation polytope) is convex and compact, and always contains the product measure, so $\mathrm{C} = 0$ is attainable. At the other end a minimum-entropy coupling gives $\mathrm{C}_{\max} = B - \min_P H(P)$, and since $H(P) \geq \max_i H(Y_i) \geq B/k$,

$$\mathrm{C}_{\max} \;\leq\; W \;=\; B - \max_i H(Y_i) \;\leq\; (1 - 1/k)\, B, \tag{3}$$

with equality throughout, $(k{-}1)H(Y_1)$, for an equal-marginal profile, attained by the comonotone coupling. Because C is continuous on a connected set, every intermediate value is realised. Minimising the concave H over a polytope attains its minimum at a vertex, so for small k we verify $\mathrm{C}_{\max}$ by exact vertex enumeration: k fair bits span exactly $[0, k{-}1]$ bits (Table 5).

A clean witness is n fair coins against n coins conditioned on exactly $n/2$ heads: identical marginals and $B = n$ bits, but $\mathrm{C} = n - \log_2 \binom{n}{n/2}$ (1.415 bits at $n{=}4$), so every marginal rule must score them identically [24].

3.1 An exact audit, and why a benchmark result cannot settle this

To score the deployed rules where C is known exactly we use the synthetic list-operation family of ParallelBench [9] and its analytic C, plus two controls of our own. Enumerated and closed-form C agree to 1.6×10^{-14} bits. Over the 355 candidate sets, the six rules collapse to four distinct rankings and the top-2 margin ranks C far better than confidence, by 0.394 $[0.296, 0.494]$ under a paired bootstrap of the difference (Table 6). The bootstrap is paired because the rules score the *same* sets.

Every proxy attains a perfect safe-set AUROC of 1.000 here, and that number is empty: on this family $\mathrm{C} = 0$ holds *if and only if* $B = 0$, so any certainty detector scores perfectly. We therefore treat the ranking above as a hypothesis to test on the four checkpoints rather than a prescription, and §7 finds it reverses.

Why this is a reference point and not a result. Eq. (3) quantifies over all joints compatible with a profile, not over the ones *a particular trained model* produces, which is what a decoder faces. §6 measures that.

4 A measurement protocol for dLLM marginals

Everything in §6 and §7 rests on one contract: given a masked canvas and a position p, return $p_\theta(Y_p \mid \text{canvas})$. Three gates establish that contract per checkpoint before any number is produced.

The estimator. C needs the joint entropy $H(Y_S \mid c)$, which is not directly available, whereas the marginal terms are. We estimate it by the chain rule along randomly sampled reveal orders. For a permutation π of S we draw one ancestral path through the model's any-order conditionals and accumulate the *exact* entropy of each predicted categorical,

$$\hat{H}_\pi(Y_S \mid c) \;=\; \sum_{t=1}^{|S|} H_\theta\big(Y_{\pi(t)} \mid y_{\pi(<t)}, c\big), \qquad y_{\pi(t)} \sim p_\theta\big(\,\cdot \mid y_{\pi(<t)}, c\big), \tag{4}$$

and average over M independent orders:

$$\widehat{\mathrm{C}}_M(S \mid c) \;=\; \underbrace{\sum_{i \in S} H_\theta(Y_i \mid c)}_{B(S \mid c)} \;-\; \frac{1}{M} \sum_{m=1}^{M} \hat{H}_{\pi_m}(Y_S \mid c). \tag{5}$$

If the conditionals compose into a joint, the chain rule makes $\hat{H}_\pi$ unbiased for $H(Y_S \mid c)$ for *any* single order, so averaging only reduces variance. We take $M{=}64$ throughout, report the per-block standard error $\mathrm{sd}_m(\hat{H}_{\pi_m})/\sqrt{M}$, and correct for it wherever $\widehat{\mathrm{C}}$ enters a variance decomposition. The estimator queries the model only through conditionals, so the identical code runs on the exact oracle of §3.1, where it recovers analytic C to 0.0087 bits, and on a checkpoint. Cost is $M|S|$ conditional queries per set, batched across orders.

What $\widehat{\mathrm{C}}$ estimates when the conditionals do not compose. Let q_π be the joint over S produced by revealing S in order π with the model's conditionals, $\bar{q} = \mathbb{E}_\pi q_\pi$ the joint of a sampler whose order is uniformly random, and p_i the parallel marginals. Each $\hat{H}_\pi$ is unbiased for $H(q_\pi)$ whatever the conditionals, so $\mathbb{E}\widehat{\mathrm{C}}_M = B - \mathbb{E}_\pi H(q_\pi)$. The cost of a parallel commit relative to these samplers obeys the exact identity

$$\mathbb{E}_\pi \,\mathrm{KL}(q_\pi \,\|\, \textstyle\prod_i p_i) \;=\; \mathrm{KL}(\bar{q} \,\|\, \textstyle\prod_i p_i) \;+\; I(\pi; Y_S), \tag{6}$$

where $I(\pi; Y_S) \le \log_2 k!$ is the information the reveal order carries about the outcome. When the conditionals compose, all three of $\mathbb{E}\widehat{\mathrm{C}}_M$, the left side and the first term on the right equal C, and $I(\pi; Y_S) = 0$. When they do not, the three separate. If every q_π keeps the parallel marginals, $\mathbb{E}\widehat{\mathrm{C}}_M$ is the left side, and it exceeds the cost relative to a random-order sampler by the order information. Two fair bits show the gap: if one order makes the second bit agree with the first with probability 0.9 and the other makes it disagree with probability 0.9, each order gives 0.531 bits while the mixture is uniform and costs nothing. If the q_π do not keep the parallel marginals, $\widehat{\mathrm{C}}$ is not a divergence at all and can be negative. We measured both terms of (6) directly on a subset (App. K). On its random blocks the per-order KL cost, the left side, exceeds $\widehat{\mathrm{C}}$ by a median of 0.4 to 1.1 bits on LLaDA-base and Dream-base prose and in their deployed regime, and $\widehat{\mathrm{C}}$ ranks blocks like it (Spearman 0.95 and 0.90, pooled across settings). The residuals of §6 are therefore residuals for $\widehat{\mathrm{C}}$, not for the KL cost. On Dream-instruct prose the gap is 4.4 and 5.6 bits.

Gate 1: which logit index carries position p? A dLLM's head need not be aligned to its own positions. Dream keeps the shift of the autoregressive model it was initialised from, so its hidden state at i predicts the token at $i{+}1$ [2], while LLaDA predicts masked tokens in place [1]. We do not take either on faith. We mask the block, then score the log-probability the model assigns to the tokens that were *actually* there, reading at index $p + \delta$ for $\delta \in \{-1, 0, +1\}$. Because the true tokens are hidden, a bidirectional model cannot win by copying: only the correct convention scores well. We require the winner to agree at $k{=}1$ and $k{=}4$ and to beat the runner-up by at least one bit.

The gate recovers each model's documented convention without being told it, by margins of 6.9 to 11.0 bits (Fig. 1a, App. B).

Gate 2: is the forward pass reproducible? Two identical calls must return identical logits. Our runs also required that scoring a sequence alone or in a batch move no argmax, although batch composition moves log-probabilities by up to 0.56 (App. B). In the released gate the two batch shapes the estimator combines must agree on every argmax and on B to within 0.074 bits RMS, a tenth of the smallest residual we report. On a second GPU all four checkpoints pass with the two shapes

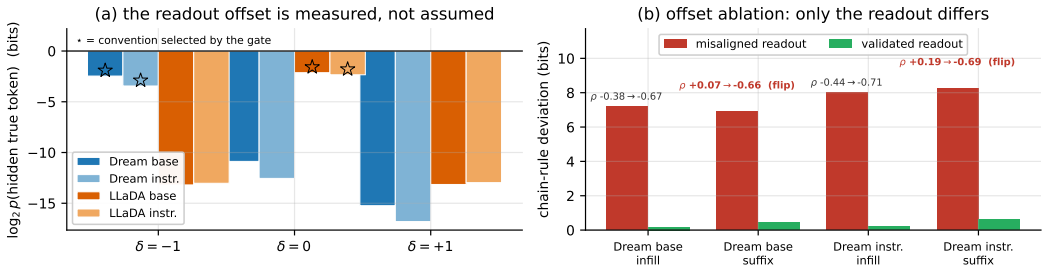


Figure 1: **(a)** The offset gate. We mask a block and score the log-probability the model assigns to the tokens that were actually there, reading at index $p + \delta$. The truth is hidden, so a bidirectional model cannot win by copying. Stars mark the convention the gate selects. Dream and LLaDA select *opposite* conventions, each matching its own documentation, without being told. **(b)** The offset ablation: identical corpus, blocks, reveal orders and code path, with only the readout changed. Chain-rule deviation [32] inflates 14 to 41× across both Dream checkpoints, and in the deployed regime the confidence-error correlation ρ is erased on Dream-base and reversed on Dream-instruct.

agreeing exactly, while scoring alone, which the measurement never does, flips 1 of 64 argmaxes on each Dream checkpoint (App. K). We also assert that the first sequentially decoded position, drawn from the parallel commit's distribution, never disagrees with it, and count violations.

Gate 3: does it write? We greedy-decode into an all-masked canvas using the calibrated offset and read the output. At a wrong offset the decode collapses on all three checkpoints we probed: one token fills 21 to 32 of 32 prose positions, against runs of at most 3 at the calibrated offset (App. K). The submitted gate only required a non-empty decode, so the released code also requires a wrong offset to produce such a collapse, and fails on any error.

5 What a wrong readout manufactures

To measure what a misaligned readout costs we hold the corpus, blocks, reveal orders and code path fixed and change only the offset, from the measured -1 to the naive 0. Three quantities move.

The consistency diagnostic. Chain-rule deviation [32], exactly zero for conditionals that compose into one joint, reads 6.9 to 8.3 bits under the misaligned readout against 0.18 to 0.59 once corrected, across both Dream checkpoints and both regimes: a factor of 14 to 41. Read naively, the misaligned number would say that these checkpoints' conditionals are far from composing into a joint.

The headline correlation. In the deployed regime the rank correlation between confidence and parallel-decoding error reads $+0.07$ $[-0.06, 0.19]$ and $+0.19$ $[0.07, 0.31]$ under the misaligned readout, against -0.66 and -0.69 once corrected. The misaligned readout erases the correlation on Dream-base and reverses it on Dream-instruct. An earlier version of this work reported these numbers and concluded that confidence is uninformative about parallel-decoding error in the deployed regime. That conclusion came from the readout.

Total correlation itself. Median $\widehat{\mathrm{C}}$ reads -4.3 and -4.8 bits under the misaligned readout in the infill regime, against $+0.78$ and $+0.05$ corrected. Since $\mathrm{C} \geq 0$ for any genuine joint, a strongly negative estimate is impossible rather than surprising, so the estimator flags the error that the disagreement metric hid.

6 How much of C the marginals explain

We apply (5) to 247 to 516 non-overlapping k=4 blocks per checkpoint and regime, plus a k=8 replication, and ask how much of the resulting variation in C a marginal-only predictor recovers. Predictors are scored out-of-fold with document-grouped folds and corrected for the meter's own variance, which is 0.7 to 7.5% of $\mathrm{Var}[\widehat{\mathrm{C}}]$. We sweep k-NN and ridge over five feature sets and keep the best out-of-fold score (nested selection moves it by at most 0.035 bits, App. I).

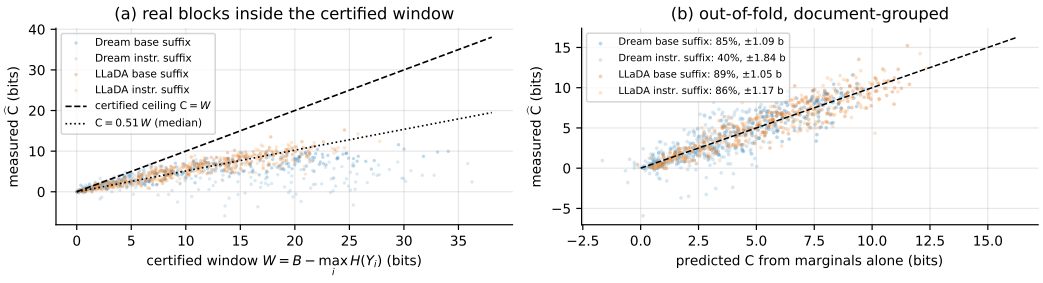


Figure 2: **Deployed (suffix) regime only.** The infill settings are in Table 1. **(a)** Measured $\widehat{\mathrm{C}}$ against the certified window $W = B - \max_i H(Y_i)$ of (3). The dashed line is the ceiling $\mathrm{C} = W$ and the dotted line the median fraction of the window outside Dream-instruct. **(b)** The headline: best marginal-only prediction of C against the truth, out-of-fold with document-grouped folds. The scatter about the diagonal is the residual left by the strongest marginal-only predictor we fit.

The marginals predict C well, mostly through the window they certify. The worst case (3) confines C to $[0, \mathrm{C}_{\max}] \subseteq [0, W]$, with median W of 1.3 to 10.8 bits across settings, and Fig. 2a shows where real blocks fall inside $[0, W]$. Outside Dream-instruct, the best marginal-only predictor leaves a residual of 0.74 to 1.17 bits (Table 1). A window width and a residual are different statistical objects, so we compare the residual with a reference that knows the marginal entropies, hence W, but nothing about the coupling, with C uniform on $[0, W]$. Knowing $\mathrm{C}_{\max}$, which top-8 profiles cannot give, would narrow it. The measured residual is 1.8 to 3.7 times below that reference. Most of this predictability is the window itself. Blocks with W above one bit sit at a median of 0.44 to 0.56 of W, a one-parameter predictor $f \cdot W$ matches the best predictor on LLaDA-base (0.74 and 1.07 bits against 0.74 and 1.05), and outside Dream-instruct, given W, the rank correlation of any of the six rules with C lies between -0.28 and $+0.23$ (Table 13). Outside Dream-instruct the full profile adds most on Dream-base in the deployed regime, where it lowers the residual from 1.81 to 1.09 bits.

Two families, one outlier, and where it comes from. Six of the eight settings agree closely. Dream-instruct is the exception (1.62 and 1.84 bits), and the exception is concentrated on prose. Under one pooled predictor its residual is 2.23 and 2.58 bits on prose, above every other checkpoint's prose residual, and 0.92 and 1.28 bits on code, its training domain (Table 12). Its chain-rule deviation on prose is the largest we measure. There the KL cost of a parallel commit exceeds $\widehat{\mathrm{C}}$ by several bits (App. K), so $\widehat{\mathrm{C}}$ does not measure it. It is not a generic instruction-tuning effect, since LLaDA-instruct sits with the base checkpoints.

Robust to block size. Repeating the measurement at k=8 on both base checkpoints gives residuals of 1.60 to 2.59 bits, 3.4 to 5.0 times below the coupling-agnostic reference, against 1.8 to 3.7 times at k=4 (Table 10).

Where $\widehat{\mathrm{C}}$ leaves $[0, W]$. No genuine joint gives $\mathrm{C} < 0$ or $\mathrm{C} > W$. Beyond two standard errors, 0% to 25% of blocks fall below zero and 0.4% to 9.5% above W. Outside Dream-instruct 79% of these are hairlines within 0.2 bits, 86% of them on code. There the per-block standard error under-covers: when the sampled paths nearly coincide it is close to zero and misses the rare branches, so even the KL version, which cannot be negative in expectation, falls below minus two standard errors on some of them (App. K). The substantive cases, beyond 0.2 bits, are 0.4% to 5.9% of blocks outside Dream-instruct and 8.1% and 14.6% on it (Table 12). We report these rates rather than clipping. Excluding the blocks below zero moves Dream-instruct's residual from 1.62/1.84 to 1.19/1.57 bits and leaves the other checkpoints essentially unchanged.

7 Proxy fidelity on real checkpoints

Table 2 scores all six marginal rules against measured C. All 48 model-regime-proxy correlations are negative and exclude zero under a document-clustered bootstrap. That much is expected from $\mathrm{C} \leq W$ once a rule tracks certainty: given W, the correlations outside Dream-instruct fall to between

Table 1: **How well the marginals predict** C**.** C is estimated per candidate set with the order-averaged chain-rule estimator, in bits. "s.d." is the block-to-block standard deviation after subtracting the estimator's own variance. "% Var" is out-of-fold R^2 with document-grouped folds, corrected for estimator variance, for a predictor built from the summed marginal entropy B alone and for the best of 40 k-NN and ridge predictors over five feature sets up to the full top-8 profile, which subsumes every *single-step* rule we score but not the cross-step stability rules of [6, 19]. The residual columns are noise-corrected residual standard deviations: *best* for that predictor (nested selection moves it by at most 0.035 bits, App. I), fW for the one-parameter predictor $f \cdot W$ with f fitted out-of-fold, and *ref.* for a coupling-agnostic reference that knows the marginal entropies, hence W, with C uniform on $[0, W]$. $W = B - \max_i H(Y_i)$ is the certified window of (3), at the median block.

		measured C		% Var[C]		residual s.d. (bits)			window
Model / regime	n	mean	s.d.	B	best	best	fW	ref.	W
Dream base / infill	494	1.70	2.25	65	75	1.13	1.41	2.04	2.0
Dream base / suffix	247	4.90	2.79	81	85	1.09	1.81	4.06	10.8
Dream instr. / infill	494	0.54	1.89	3	26	1.62	2.03	2.53	1.3
Dream instr. / suffix	247	2.54	2.37	15	40	1.84	3.59	4.01	6.7
LLaDA base / infill	516	1.78	2.38	86	90	0.74	0.74	1.52	1.7
LLaDA base / suffix	258	5.47	3.24	88	89	1.05	1.07	3.48	10.3
LLaDA instr. / infill	516	1.67	2.28	78	85	0.87	0.95	1.60	1.6
LLaDA instr. / suffix	258	5.03	3.13	81	86	1.17	1.34	3.44	9.5

-0.28 and $+0.23$ (Table 13). The informative part is how the rules compare with each other, and two comparisons run against the synthetic audit.

Aggregation matters more than the statistic. In every one of the eight settings, pooling across the block beats taking its weakest position: confidence-mean beats confidence-min, margin-mean beats margin-min, and summed neg-entropy beats maximum neg-entropy, by 0.066 to 0.235 in $|\rho|$, and a paired bootstrap excludes zero in all 24 comparisons (Table 13). The "commit only if the least confident member is confident" rule that several systems use is the aggregation that throws the most away.

The synthetic prescription does not transfer. On our exact tasks the top-2 margin ranks C far better than confidence, by 0.394 $[0.296, 0.494]$ under a paired bootstrap (Table 6). On real checkpoints the min versions reverse it: confidence is better in seven of eight settings, significantly so in six, by up to 0.189 $[0.121, 0.264]$. For the pooled versions neither dominates. Mean confidence is significantly better in five settings, mean margin in the two Dream-instruct settings, and one is a tie, and summed entropy against mean confidence splits three, two and three (Table 13). The pooled rules reach $\rho = -0.83$ to -0.94 outside Dream-instruct. These are statements about predicting C. Whether a pooled rule makes a better decoder is a separate question that needs an end-to-end comparison.

The usual outcome variable is a weak stand-in. Parallel-commit error is normally measured as disagreement between a parallel commit and a sequential reference decode. Against measured C that variable correlates at only $+0.23$ to $+0.64$. It is also not always well defined: in the deployed regime a greedy reference decoded under 8 different reveal orders yields a median of 2 distinct assignments, with pairwise disagreement 0.21 to 0.31, so the reference against which error is measured depends on an arbitrary choice. In the infill regime it is order-invariant (1 distinct assignment, pairwise disagreement 0). This is our reason for measuring C rather than proxying it. Every checkpoint reacts to what is filled in, with median total-variation distance 0.31 to 0.37 in the deployed regime and 0.05 to 0.08 in infill (Table 7), and the structural invariant of §4 holds on all 3,030 blocks.

8 Limitations

What C **means for these models.** Eq. (1) to (3) and the reading of C as the KL cost of a parallel commit assume that the conditionals compose into one joint, and a dLLM's any-order conditionals need not [33]. The estimator does not need that assumption. $\widehat{\mathrm{C}}$ is always B minus an order-averaged chain-rule entropy. When each reveal order keeps the parallel marginals, Eq. (6) relates it to the cost of a parallel commit, and otherwise it is not a divergence (§4). What is lost without a joint is a single quantity that deserves the name total correlation. Numbers measured on one GPU type need not

Table 2: **Every marginal rule against measured C, on real checkpoints.** Spearman ρ between each proxy's safety score and the meter's $\widehat{\mathrm{C}}$, so a faithful proxy is negative. * marks intervals excluding zero under a document-clustered bootstrap (10^4 resamples), and every one of the 48 correlations does. Pooling across the block (mean, sum) beats gating on its weakest position (min, max) in every setting. The last column is a *paired* bootstrap of the difference in ρ between the min versions of top-2 margin and confidence on the same blocks, positive where confidence is the better rule. The pooled comparisons are in Table 13. This is the correct test for "does margin beat confidence" and is strictly stronger than asking whether two marginal intervals overlap. Our exact audit (Table 6) favours margin, and on real checkpoints that reverses.

Model / regime	conf (min)	conf (mean)	marg (min)	marg (mean)	negH (max)	negH (sum)	marg$-$conf
Dream base / infill	-0.72^*	-0.83^*	-0.69^*	-0.83^*	-0.77^*	-0.83^*	$+0.033^*$
Dream base / suffix	-0.79^*	-0.89^*	-0.65^*	-0.86^*	-0.78^*	-0.88^*	$+0.140^*$
Dream instr. / infill	-0.23^*	-0.33^*	-0.28^*	-0.37^*	-0.17^*	-0.23^*	-0.045^*
Dream instr. / suffix	-0.46^*	-0.54^*	-0.45^*	-0.58^*	-0.36^*	-0.45^*	$+0.007$
LLaDA base / infill	-0.73^*	-0.86^*	-0.69^*	-0.85^*	-0.79^*	-0.89^*	$+0.041^*$
LLaDA base / suffix	-0.78^*	-0.89^*	-0.60^*	-0.83^*	-0.84^*	-0.94^*	$+0.189^*$
LLaDA instr. / infill	-0.75^*	-0.85^*	-0.71^*	-0.84^*	-0.80^*	-0.87^*	$+0.041^*$
LLaDA instr. / suffix	-0.76^*	-0.89^*	-0.61^*	-0.85^*	-0.81^*	-0.90^*	$+0.144^*$

reproduce bit for bit on another: on identical blocks B differs between the A100 and the GB10 by a mean of 0.04 to 0.19 bits per setting and by up to 3.5 bits on single Dream blocks.

What the residual bounds, and what it does not. Our predictor reads per-position top-8 probabilities and entropies. That subsumes every single-step rule we score, at any aggregation, but not the full marginal and not the cross-step stability rules of [6, 19], which read more than one decoding step. Within the class we fit, a better predictor could only move the residual down, so we state it as a bound and not an impossibility. Two things could undercut that and neither does. The winner is chosen over 40 candidates on the same out-of-fold scores we report, which biases the residual downward, so we redo the analysis selecting on inner folds and scoring on outer ones, and it moves by at most 0.035 bits (App. I). Quadrupling M to 256 on LLaDA-base moves the residual by at most 0.034 bits (App. G), and the estimator recovers analytic C on the oracle.

Scope. $k{=}4$ blocks with $k{=}8$ as a check, greedy decoding, one 128-token window per document, English prose and Python (App. A). These numbers describe one decoding step, not end-to-end generation quality, and they say nothing about which set a decoder should commit, since we score fixed contiguous blocks rather than the sets a rule would choose, and a deployed decoder may commit larger sets and sample rather than take the argmax. Four checkpoints across two families, run on raw text without chat formatting, which keeps inputs comparable but is not how an instruct model is deployed.

9 Conclusion

A rule reading per-position marginals cannot determine the dependency of a parallel commit, only bound it by $\mathrm{C}_{\max} \le W$. On four checkpoints from two families and two regimes, the marginals nevertheless predict the estimate $\widehat{\mathrm{C}}$ with a residual of 0.74 to 1.17 bits outside Dream-instruct, 1.8 to 3.7 times below a reference that ignores the coupling, mostly because real blocks use a stable fraction of their window. $\widehat{\mathrm{C}}$ ranks blocks like the KL cost of a parallel commit but sits below it on prose and in the deployed regime, and on Dream-instruct prose it does not measure that cost. What a dependency-aware decoder gains over a marginal rule is a different quantity, which needs a matched-compute decoding study.

Two widely used open dLLMs require opposite logit-position conventions. The wrong one collapses greedy prose on both families, yet a pipeline that never decodes text does not notice. The cheapest defence is to check the readout against something the model cannot see, and to report the check.

References

[1] Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, et al. Large Language Diffusion Models. arXiv:2502.09992, 2025.

[2] Jiacheng Ye, Zhihui Xie, Lin Zheng, Jiahui Gao, et al. Dream 7B: Diffusion Large Language Models. arXiv:2508.15487, 2025.

[3] Zhihui Xie, Jiacheng Ye, Lin Zheng, Jiahui Gao, et al. Dream-Coder 7B: An Open Diffusion Language Model for Code. arXiv:2509.01142, 2025. (Checkpoints: Dream-Coder-v0 Base / Instruct 7B.)

[4] Chengyue Wu, Hao Zhang, Shuchen Xue, Zhijian Liu, et al. Fast-dLLM: Training-free Acceleration of Diffusion LLM by Enabling KV Cache and Parallel Decoding. arXiv:2505.22618, 2025.

[5] Tianyi Wu, Xiaoxi Sun, Yanhua Jiao, Yulin Li, et al. Dynamic-dLLM: Dynamic Cache-Budget and Adaptive Parallel Decoding for Training-Free Acceleration of Diffusion LLM. arXiv:2606.26120, 2026.

[6] Amr Mohamed, Guokan Shang, and Michalis Vazirgiannis. LESS Is More: Mutual-Stability Sampling for Diffusion Language Models. arXiv:2606.16908, 2026.

[7] Bumjun Kim, Dongjae Jeon, Moongyu Jeon, and Albert No. DAPD: Dependency-Aware Parallel Decoding via Attention for Diffusion LLMs. In ICML, 2026. arXiv:2603.12996.

[8] Lipeng Wan, Junjie Ma, Jianhui Gu, Zeyang Liu, et al. GeoBlock: Inferring Block Granularity from Dependency Geometry in Diffusion Language Models. arXiv:2603.26675, 2026.

[9] Wonjun Kang, Kevin Galim, Seunghyuk Oh, Minjae Lee, et al. ParallelBench: Understanding the Trade-offs of Parallel Decoding in Diffusion LLMs. In ICLR, 2026. arXiv:2510.04767.

[10] Anji Liu, Oliver Broadrick, Mathias Niepert, and Guy Van den Broeck. Discrete Copula Diffusion. arXiv:2410.01949, 2024.

[11] Shaorong Zhang, Longxuan Yu, Rob Brekelmans, Luhan Tang, et al. Generation Order and Parallel Decoding in Masked Diffusion Models: An Information-Theoretic Perspective. arXiv:2602.00286, 2026.

[12] Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, et al. Simple and Effective Masked Diffusion Language Models. In NeurIPS, 2024. arXiv:2406.07524.

[13] Jiaxin Shi, Kehang Han, Zhe Wang, Arnaud Doucet, et al. Simplified and Generalized Masked Diffusion for Discrete Data. In NeurIPS, 2024. arXiv:2406.04329.

[14] Guhao Feng, Yihan Geng, Jian Guan, Wei Wu, et al. Theoretical Benefit and Limitation of Diffusion Language Model. arXiv:2502.09622, 2025.

[15] DiffusionGemma Team. DiffusionGemma Technical Report. arXiv:2608.00146, 2026. (Checkpoint: diffusiongemma-26B-A4B-it.)

[16] Sitan Chen, Kevin Cong, and Jerry Li. Optimal Inference Schedules for Masked Diffusion Models. arXiv:2511.04647, 2025.

[17] Fei Huang, Tianhua Tao, Hao Zhou, Lei Li, et al. On the Learning of Non-Autoregressive Transformers. In ICML, 2022. arXiv:2206.05975.

[18] Kecheng Chen, Ziru Liu, Xijia Tao, Hui Liu, et al. Beyond Confidence: Adaptive and Coherent Decoding for Diffusion Language Models. arXiv:2512.02044, 2025.

[19] Yue Wu and Jian Huang. Stability-Weighted Decoding for Diffusion Language Models. arXiv:2604.17068, 2026.

[20] Lin Yao. Don't Commit Alone: Joint Token Commitment in Diffusion Large Language Models. arXiv:2607.04469, 2026.

[21] Jai Sharma, Yifan Wang, and Bryan Li. Neural Estimation of Pairwise Mutual Information in Masked Discrete Sequence Models. arXiv:2605.20187, 2026.

[22] Liran Ringel, Ameen Ali, and Yaniv Romano. Dependency-Guided Parallel Decoding in Discrete Diffusion Language Models. arXiv:2604.02560, 2026.

[23] Linjun Li. One-step lowest-variance selection in a Gaussian random-field model motivated by masked diffusion: Total correlation and a square root collision threshold. arXiv:2607.17522, 2026.

[24] Nazia Riasat. When Marginals Match but Structure Fails: Covariance Fidelity in Generative Models. Extended version of a paper in MathAI, 2026. arXiv:2603.17041.

[25] Ali Asaria, Tony Salomone, and Deep Gandhi. Neither Parallel Nor Sequential: How DiffusionGemma Actually Commits Tokens. arXiv:2606.14620, 2026.

[26] Yunxiao Zhao and Changxiao Cai. Adaptation to Intrinsic Dependence in Diffusion Language Models. arXiv:2602.20126, 2026.

[27] Lizhuo Luo, Zhuoran Shi, Jiajun Luo, Zhi Wang, et al. DAWN: Dependency-Aware Fast Inference for Diffusion LLMs. arXiv:2602.06953, 2026.

[28] Xueyu Zhou, Yangrong Hu, and Jian Huang. DOS: Dependency-Oriented Sampler for Masked Diffusion Language Models. arXiv:2603.15340, 2026.

[29] Heqiang Qi, Wei Huang, Mingyuan Bai, and Xiangming Meng. Cluster-Level Attention-Guided Parallel Decoding for Masked Diffusion Language Models. arXiv:2605.29607, 2026.

[30] Tamim Zoabi, Ameen Ali, Liran Ringel, and Lior Wolf. Mean-Field Parallel Decoding for Discrete Diffusion Language Models. arXiv:2606.15805, 2026.

[31] Bohang Sun, Max Zhu, Francesco Caso, Jindong Gu, et al. The Path Matters: Learning a Token-Commitment Policy for Diffusion Language Models. arXiv:2605.24697, 2026.

[32] Lin Yao. Decoding in Order-Agnostic Language Models: Chain-Rule Deviation and Uniform Spreading. arXiv:2606.00997, 2026.

[33] Tom Young, Yunan Chen, and Yang You. Inconsistencies in Masked Language Models. arXiv:2301.00068, 2022.

[34] Suvadip Sana, Sami Wolf, Neer Mehta, and Alina Shah. Mixing Times of Glauber Dynamics on Masked Language Models. arXiv:2605.16378, 2026.

[35] Jeonseong Kim. Path-Dependent Denoising: A Non-Conservative Field Perspective on Order Collapse in Diffusion Language Models. arXiv:2605.09303, 2026.

[36] Iskander Azangulov, Teodora Pandeva, Niranjani Prasad, and Javier Zazo. Parallel Sampling from Masked Diffusion Models via Conditional Independence Testing. In ICLR, 2026. arXiv:2510.21961.

[37] Michael Hersche, Nicolas Menet, Ronan Tanios, and Abbas Rahimi. Locally Coherent Parallel Decoding in Diffusion Language Models. arXiv:2603.20216, 2026.

A Experimental details

Corpus. 285 documents: 121 WikiText-103 articles (`wikitext-103-raw-v1`, those with at least 600 characters of body text) for English prose, and all 164 OpenAI HumanEval problems for Python, which is the natural code domain for Dream-Coder. We take the first 128 tokens of each document under the checkpoint's own tokenizer as one window. Documents shorter than 128 tokens under a checkpoint's tokenizer are dropped, so Dream sees all 121 prose documents and 126 of the code problems (247 windows) and LLaDA sees 121 and 137 (258 windows).

The two regimes. A block is k consecutive positions inside a window. *Suffix* masks everything from the block to the end of the window, which is the deployed setting where a decoder faces a prompt and an otherwise empty canvas. One suffix block is taken per window, starting at the midpoint. *Infill* masks only the block itself, so the model sees full right context. Up to two infill blocks are taken per window, at starts drawn without replacement from the window's second quarter onward and spaced $4k$ apart. Blocks are shuffled under seed 0. The cap of 600 per regime never binds, so suffix has one block per window and infill two, which gives 247 or 258 suffix blocks and 494 or 516 infill blocks.

Estimator and diagnostics. $M{=}64$ reveal orders for $\widehat{\mathrm{C}}$, 8 orders for the chain-rule deviation, 8 reference orders for the order-sensitivity control, and the top 8 probabilities per position for the profile features. Reference decodes are greedy. The per-block standard error is $\mathrm{sd}_m(\hat{H}_{\pi_m})/\sqrt{M}$.

Variance correction. The meter is noisy, so $\mathrm{Var}[\widehat{\mathrm{C}}] = \mathrm{Var}[\mathrm{C}] + \mathbb{E}[\mathrm{se}^2]$, and the same offset appears in any out-of-fold squared error. We subtract the measured $\mathbb{E}[\mathrm{se}^2]$ from both before forming an R^2 or a residual, which turns statements about $\widehat{\mathrm{C}}$ into statements about C. A block counts as significantly negative when $\widehat{\mathrm{C}} < -2\,\mathrm{se}$.

Predictors. Five document-grouped folds under seed 0, so every block of a document falls in the same fold and no document is split across train and test. Features are standardised. The sweep is k-NN with $k \in \{5, 10, 20, 40\}$, used when $k < n/6$, and ridge with $\lambda \in \{0.1, 1, 10, 100\}$ and an unpenalised intercept, over the five feature sets of Table 1. App. I repeats it with the selection nested inside inner folds.

Compute. One A100 80GB. Both families run in `bfloat16` under `transformers` 4.46.2, which Dream's remote code requires. A full measurement run is roughly 4,700 batched forward calls per checkpoint. Everything in §3 and App. E is exact and runs on CPU.

Availability. Code, configurations, per-block results and the verification probe are at `https://github.com/DovudAsadov/dllm-total-correlation`, tag `paper-numbers`.

B Gate outputs

Table 3 gives gate 1 for every checkpoint: the measured readout offset against the documented one, the evidence at each δ, and the margin over the runner-up. Table 4 gives gate 2 in full, namely exact reproducibility across repeat calls and equivalence between scoring a sequence alone and inside a batch, together with the gate 3 decode.

C Why DiffusionGemma is not in this study

An earlier version included DiffusionGemma-26B [15], whose commit behaviour has been instrumented elsewhere [25]. It does not belong: its technical report states that its denoising "uses multinomial (or uniform) diffusion rather than masked diffusion", it initialises its canvas with uniformly random tokens, and its configuration declares no mask token. There is no masked position to commit and no marginal-at-a-mask to read, so this paper's object is undefined for it.

Table 3: **The readout convention is measured, not assumed.** We mask a $k{=}4$ block and score the log-probability the model gives the tokens that were actually there, reading at index $p{+}\delta$. The true tokens are hidden, so a bidirectional model cannot score well by copying, and only the correct convention wins. The gate recovers each checkpoint's documented convention without being told it, and the two families are *opposite*: Dream keeps the autoregressive shift of the backbone it was initialised from, LLaDA predicts masked tokens in place. The two Dream rows are the Dream-Coder checkpoints [3], adapted from an autoregressive backbone by the recipe of [2], and the gate finds they keep its shift. "gap" is the winning δ's margin over the runner-up in bits, at $k{=}4$ (the block size of the three columns before it) and with one hidden token ($k{=}1$). The gate requires both to exceed one bit. "flips" counts argmax changes between two identical forward calls. Any nonzero value would be a noise floor that correlates with confidence and therefore contaminates a proxy audit. Reading Dream at $\delta{=}0$, the convention our own earlier version used, costs 6.9 bits of log-probability on the correct token at $k{=}1$ and 8.4 at $k{=}4$, a factor of $\sim$120 or more in probability (§5).

	readout offset		$\log_2 p$(hidden true token) read at $p{+}\delta$			gap (bits)			
Model	measured	documented	$\delta{=}{-}1$	$\delta{=}0$	$\delta{=}{+}1$	$k{=}4$	$k{=}1$	flips	gates
Dream-Coder-7B (base)	-1	-1 (AR shift)	-2.47	-10.87	-15.24	8.4	6.9	0	✓
Dream-Coder-7B (instruct)	-1	-1 (AR shift)	-3.44	-12.55	-16.80	9.1	8.4	0	✓
LLaDA-8B (base)	0	0 (in place)	-13.19	-2.13	-13.15	11.0	16.0	0	✓
LLaDA-8B (instruct)	0	0 (in place)	-13.03	-2.34	-12.97	10.6	14.7	0	✓

Table 4: **Gate 2 and gate 3, in full.** Table 3 reports gate 1. *repeat*: maximum absolute difference in log-probability between two identical forward calls, and the argmax flips that difference causes. *batch*: the same comparison between scoring a sequence alone and scoring it inside a batch. Exact reproducibility on repeat calls is what the proxy audit needs, because a nondeterministic forward flips argmaxes preferentially where confidence is low and so manufactures the negative confidence-error correlation the audit is trying to detect. The gate requires identical repeat calls and no alone-versus-batch argmax flip. It does not bound the alone-versus-batch change in log-probability, which is nonzero, and that change reaches the entropies the estimator uses (App. K). Repeat calls compare 8 sequences (32 positions), the alone-versus-batch column one sequence (4 positions). Gate 3 is the first tokens greedy-decoded into an all-masked canvas using the calibrated offset, truncated for width.

	repeat call		alone vs batch		gate 3
Model	max Δ	flips	max Δ	flips	greedy decode
Dream base	0	0	0.162	0	`Theatre . In 2002 , he appeared in the ...`
Dream instr.	0	0	0.556	0	`Theatre . He also appeared in the film ...`
LLaDA base	0	0	0.281	0	`Theatre . He has also appeared in the ...`
LLaDA instr.	0	0	0.224	0	`Theatre in London . He has also appear ...`

D The worst-case fibre

Table 5 tabulates the width $\mathrm{C}_{\max}$ of (3) at representative marginal profiles, by exact vertex enumeration for k fair bits.

E The exact audit

Table 6 reports the six marginal rules scored against analytic C on the synthetic family of §3.1, where enumerated and closed-form C agree to 1.6×10^{-14} bits. It is the ranking that §7 finds does not transfer to real checkpoints.

F Controls

Table 7 reports the per-setting diagnostics referenced in §7: the chain-rule deviation, the conditional-sensitivity check that distinguishes a composable joint from a degenerate one, the reference-order sensitivity of the usual outcome variable, and the structural invariant.

Table 5: **What fixing the marginals leaves undetermined.** Since $H(Y_S \mid c) \geq 0$, Eq. (1) certifies $\mathrm{C} \leq B := \sum_{i \in S} H(Y_i \mid c)$, which is exactly minus the `neg_entropy_sum` proxy. Marginal rules bound C, they do not locate it. The fibre of joints with these marginals is convex, so C attains every value in $[\mathrm{C}_{\min}, \mathrm{C}_{\max}]$: the product measure gives $\mathrm{C}_{\min} = 0$, and a minimum-entropy coupling gives $\mathrm{C}_{\max} = B - \min_P H(P)$, which for equal marginals is $(k-1)H(Y_1)$, so k fair bits span exactly $[0, k-1]$ bits (rows (a), by exact vertex enumeration). Every entry is a statement about the whole fibre of joints compatible with a marginal profile, so it bounds what the marginals fail to say and not what any particular trained model does. What a real checkpoint actually does is measured in §6.

Marginal profile	k	$B = \sum_i H(Y_i)$	$\mathrm{C}_{\min}$	$\mathrm{C}_{\max}$	diameter
(a) fair bits: exact vertex enumeration of the fibre					
2 fair bits ($c = 0.5$)	2	2.000	0.000	1.000	1.000
3 fair bits ($c = 0.5$)	3	3.000	0.000	2.000	2.000
4 fair bits ($c = 0.5$)	4	4.000	0.000	3.000	3.000
(b) $k = 4$, every position Bernoulli(c) at confidence c					
$c = 0.5$	4	4.000	0.000	3.000	3.000
$c = 0.75$	4	3.245	0.000	2.434	2.434
$c = 0.9$	4	1.876	0.000	1.407	1.407
$c = 0.95$	4	1.146	0.000	0.859	0.859
$c = 0.99$	4	0.323	0.000	0.242	0.242

Table 6: **The exact audit.** Six marginal rules scored against analytic C on all 355 candidate sets from REPLACE_RANDOM and SHUFFLE, of which 34 have $\mathrm{C} = 0$. The rules collapse to 4 distinct rankings, so one row is shown per ranking. The perfect safe-set AUROC is an identity, not a finding: here $\mathrm{C} = 0$ *if and only if* the summed marginal entropy is 0 (all 34 at 0 bits, all 321 others at ≥ 1.62), so any certainty detector scores 1.000 while measuring no dependency. *gap* is the matched-marginal score difference, exactly 0 while C differs by 1.42 bits. Comparisons use a paired bootstrap of the *difference* in Spearman on the same sets: margin beats confidence by 0.394 [0.296, 0.494]. §7 finds this reverses on the four checkpoints.

Rule	Spearman$(\cdot, \mathrm{C})$ [95% CI]	Kendall	$\mathrm{AUROC}_{\mathrm{safe}}$	gap
confidence (min), neg-entropy (max)	-0.203 $[-0.345, -0.055]$	-0.104	1.0000	0
confidence (mean)	-0.201 $[-0.344, -0.054]$	-0.103	1.0000	0
margin (min/mean)	-0.596 $[-0.683, -0.499]$	-0.539	1.0000	0
neg-entropy (sum)	-0.335 $[-0.471, -0.185]$	-0.198	1.0000	0

Table 7: **Controls.** *chain-rule dev.*: spread over reveal orders of the chain-rule log-probability at a fixed assignment, exactly zero for conditionals that compose into one joint [32]. *cond. sens.*: mean total-variation distance between $p(Y_j \mid c)$ and $p(Y_j \mid y_i, c)$. A model that ignores what has been filled in sits near zero, would trivially score as "consistent", and would trivially have $\mathrm{C} = 0$, so this separates a genuinely composable joint from a degenerate one. *distinct y*: number of distinct greedy assignments obtained under 8 different reveal orders, out of 8. When this exceeds one, the sequential "reference" decode against which parallel-commit error is usually measured is itself order-dependent, and *pairwise dis.* reports how far apart those references are. *inv. ok*: fraction of blocks satisfying the structural invariant that the first sequentially decoded position, drawn from the same distribution as the parallel commit, agrees with it. Anything below 100 is nondeterminism in the forward pass.

Model / regime	med. conf.	B	$\widehat{\mathrm{C}}$	chain-rule dev.	cond. sens.	distinct y	pairwise dis.	inv. ok %
Dream base / infill	0.538	4.2	0.78	0.18	0.080	1.0	0.00	100.0
Dream base / suffix	0.159	17.3	5.06	0.42	0.358	2.0	0.29	100.0
Dream instr. / infill	0.575	4.3	0.05	0.20	0.068	1.0	0.00	100.0
Dream instr. / suffix	0.262	11.7	2.24	0.59	0.311	2.0	0.31	100.0
LLaDA base / infill	0.565	3.6	0.68	0.11	0.050	1.0	0.00	100.0
LLaDA base / suffix	0.177	16.6	5.50	0.28	0.366	2.0	0.21	100.0
LLaDA instr. / infill	0.597	3.5	0.55	0.10	0.049	1.0	0.00	100.0
LLaDA instr. / suffix	0.205	14.8	5.03	0.37	0.352	2.0	0.21	100.0

G Estimator budget: convergence and sensitivity in M

Table 8 reports the estimator's convergence in the number of sampled reveal orders M, on a fixed set of 48 deployed-regime blocks per base checkpoint. Each smaller M reuses the first M of the same

256 orders, which the caption accounts for. Table 9 asks the separate question of whether the budget changes the answer, and repeats the analysis at $M{=}256$ on the same blocks as the main run.

Table 8: **The meter converges as $1/\sqrt{M}$.** Per-block standard error of $\widehat{\mathrm{C}}_M$ against the number of sampled reveal orders M, on a fixed block set per checkpoint. Each smaller M uses the first M of the same 256 orders, so the deviation from $\widehat{\mathrm{C}}_{256}$ in the last columns understates the error of an independent run of size M, by a factor of $\sqrt{256/(256-M)}$. The last three columns give the mean and maximum absolute deviation of $\widehat{\mathrm{C}}_M$ from $\widehat{\mathrm{C}}_{256}$ on the same blocks, and the forward-pass cost per block. The main runs use $M{=}64$, the row where the deviation from a $4\times$ larger budget is already a fraction of the residuals reported in Table 1.

M	s.e. (bits)	s.e.$\times\sqrt{M}$	mean $\lvert\widehat{\mathrm{C}}_M - \widehat{\mathrm{C}}_{256}\rvert$	max	fwd/block
Dream base, deployed regime, $n = 48$ blocks, $k = 4$					
4	2.821	5.64	2.00	10.65	16
8	1.856	5.25	1.34	4.05	32
16	1.323	5.29	1.05	3.82	64
32	0.882	4.99	0.67	1.67	128
64	0.671	5.36	0.39	1.45	256
128	0.479	5.42	0.22	0.57	512
256	0.348	5.56	0	0	1024
LLaDA base, deployed regime, $n = 48$ blocks, $k = 4$					
4	1.551	3.10	1.25	7.63	16
8	1.121	3.17	0.98	4.46	32
16	0.783	3.13	0.50	2.09	64
32	0.586	3.31	0.44	1.49	128
64	0.432	3.45	0.31	1.15	256
128	0.302	3.41	0.14	0.64	512
256	0.208	3.33	0	0	1024

Table 9: **The estimator budget does not carry the headline.** Quadrupling the number of reveal orders from $M{=}64$ to $M{=}256$ on the *same* blocks. The per-block standard error halves, as $1/\sqrt{M}$ predicts, and the reported residual moves by at most 0.034 bits. Note this is a different experiment from Table 8, which traces the convergence curve on a smaller fixed set of 48 blocks, so the standard errors are not directly comparable between the two tables. *rank agr.*: Spearman correlation between the two estimates of C across blocks, i.e. whether the two budgets order the blocks the same way. The two runs also differ in batch composition, 8 blocks per batch at $M{=}64$ and 4 at $M{=}256$, so this tests the budget and the batch shape together.

		meter s.e. (bits)		residual (bits)			
Model / regime	n	$M{=}64$	$M{=}256$	$M{=}64$	$M{=}256$	$\lvert\Delta\rvert$	rank agr.
LLaDA base / suffix	258	0.335	0.168	1.055	1.088	0.034	0.991
LLaDA base / infill	516	0.072	0.040	0.742	0.732	0.010	0.988

H Block size: the $k{=}8$ replication

Table 10 gives the per-setting numbers that §6 quotes as ranges, for blocks of eight positions on the two base checkpoints.

I Selecting the predictor without a winner's curse

Table 11 repeats the analysis of §6 with the model and feature-set selection nested inside its own folds, so no block's outcome influences the predictor that is later scored on it. This is the check that the residual is not an artifact of choosing the winner on the same numbers we report.

J Correction analyses

Table 12 splits the residual and the ill-defined blocks by domain, and Table 13 gives the paired comparisons behind §7 and the partial correlations behind §6.

Table 10: **The k=8 replication, per setting.** §6 quotes these as ranges. Same corpus, same protocol, same estimator, blocks of eight positions instead of four, on the two base checkpoints. The last two columns give the median certified window $W = B - \max_i H(Y_i)$ and the residual of the coupling-agnostic reference of Table 1 at k=8, which knows the marginal entropies but not the coupling. The k=4 column repeats Table 1 for comparison on the same checkpoint and regime.

		$\widehat{\mathrm{C}}$	% Var	unexpl. (bits)		window	ref.
Model / regime	n	mean	expl.	k=8	k=4	W	(bits)
Dream base / infill	250	8.36	90	2.19	1.13	14.3	7.37
Dream base / suffix	247	15.97	83	2.59	1.09	40.3	12.09
LLaDA base / infill	250	9.31	96	1.60	0.74	14.5	6.12
LLaDA base / suffix	250	18.29	92	2.14	1.05	36.2	10.64

Table 11: **What selecting the predictor costs.** Every prediction behind Table 1 is out-of-fold, but the winning feature set and model are chosen over 40 candidates using those same out-of-fold scores, which biases the residual downward. That bias runs the wrong way for an upper bound, so we price it rather than assume it away: *nested* repeats the whole procedure with 5 outer document-grouped folds for evaluation and 5 inner folds for selection, so no block's outcome ever influences the model that predicts it. The largest movement across the eight settings is $+0.035$ bits, and the range of residuals is unchanged at two decimal places. *selected*: the feature set and model the inner folds chose most often.

		residual (bits)			
Model / regime	n	reported	nested	Δ	selected
Dream base / infill	494	1.129	1.132	$+0.003$	`profile_plus_scalars/ridge100`
Dream base / suffix	247	1.090	1.095	$+0.005$	`per_position_conf_entropy/ridge1`
Dream instr. / infill	494	1.623	1.633	$+0.010$	`per_position_conf_entropy/ridge10`
Dream instr. / suffix	247	1.840	1.845	$+0.005$	`per_position_conf_entropy/ridge1`
LLaDA base / infill	516	0.742	0.743	$+0.000$	`per_position_conf_entropy/ridge10`
LLaDA base / suffix	258	1.055	1.072	$+0.017$	`profile_plus_scalars/ridge10`
LLaDA instr. / infill	516	0.873	0.908	$+0.035$	`per_position_conf_entropy/knn10`
LLaDA instr. / suffix	258	1.174	1.172	-0.002	`per_position_conf_entropy/ridge10`

Table 12: **Where the residual and the ill-defined blocks are.** *residual*: noise-corrected residual of one pooled ridge predictor (profile, confidence and B, λ=10) fitted on both domains and scored separately on each, with a document-clustered 95% interval. *chain-rule dev.*: median spread over reveal orders of the chain-rule log-probability at a fixed assignment [32]. $\widehat{\mathrm{C}} < 0$ and $\widehat{\mathrm{C}} > W$: percentage of blocks beyond two standard errors outside $[0, W]$, $W = B - \max_i H(Y_i)$, which no joint admits. *subst.*: the part of those beyond 0.2 bits. The rest are hairlines on near-deterministic blocks.

	residual (bits)		chain-rule dev.		impossible (% of blocks)		
Model / regime	prose	code	prose	code	$\widehat{\mathrm{C}} < 0$	$\widehat{\mathrm{C}} > W$	subst.
Dream base / infill	1.35 [1.17, 1.54]	1.01 [0.74, 1.27]	0.42	0.06	9.5	7.9	5.9
Dream base / suffix	1.33 [1.12, 1.54]	1.01 [0.86, 1.15]	0.76	0.28	0.0	1.6	1.2
Dream instr. / infill	2.23 [1.95, 2.49]	0.92 [0.54, 1.35]	0.59	0.03	25.1	8.9	14.6
Dream instr. / suffix	2.58 [2.21, 2.92]	1.28 [1.06, 1.47]	1.17	0.30	4.9	8.1	8.1
LLaDA base / infill	1.00 [0.84, 1.16]	0.54 [0.44, 0.65]	0.28	0.04	8.5	4.3	1.4
LLaDA base / suffix	1.18 [0.97, 1.38]	0.93 [0.83, 1.05]	0.47	0.21	0.0	0.4	0.4
LLaDA instr. / infill	1.16 [0.98, 1.35]	0.58 [0.47, 0.68]	0.27	0.04	7.6	9.5	2.3
LLaDA instr. / suffix	1.45 [1.22, 1.68]	1.00 [0.83, 1.17]	0.67	0.20	0.0	0.8	0.4

K Verification on a second GPU

We reran the estimator on a subset of blocks with `scripts/verify/probe.py` on an NVIDIA GB10, with the checkpoints and blocks of the main run, to measure both terms of (6) and to test the gates where the main run did not. For each of 32 sampled assignments per block we score all 4! reveal orders from the 14 partially revealed canvases, which gives $I(\pi; Y_S)$ up to Monte Carlo error, and we score the samples under the parallel marginals, which gives the KL version. Table 14 gives medians over all probed blocks, which oversample flagged ones, and over the random blocks alone.

Table 13: **Paired comparisons, and what the rules see beyond certainty.** The first three columns are paired, document-clustered bootstraps (2,000 resamples) of $\rho(\text{first rule}, \widehat{\mathrm{C}}) - \rho(\text{second rule}, \widehat{\mathrm{C}})$ on the same blocks, positive where the second rule is the better one. * marks intervals excluding zero. *pooled vs min*: the largest upper end, over the three statistics, of the interval for pooled minus min. Negative throughout means pooling wins every one of the 24 paired comparisons. The last two columns are the smallest and largest partial Spearman correlation, over the six rules, of a rule with $\widehat{\mathrm{C}}$ given the window W (per-rule values and intervals are in `results/e5v3_corrections.json`). Without conditioning the pooled rules reach ρ of -0.83 to -0.94 outside Dream-instruct (Table 2), so most of what a rule knows about C is the width of the window it certifies.

	marg−conf	marg−conf	negH−conf	pooled vs	partial ρ given W	
Model / regime	(min)	(mean)	(sum vs mean)	min (max hi)	over the six rules	
Dream base / infill	$+0.033^*$	$+0.004$	-0.000	-0.048	-0.14	$+0.04$
Dream base / suffix	$+0.140^*$	$+0.037^*$	$+0.017$	-0.063	-0.28	$+0.21$
Dream instr. / infill	-0.045^*	-0.036^*	$+0.101^*$	-0.044	-0.21	$+0.28$
Dream instr. / suffix	$+0.007$	-0.032^*	$+0.094^*$	-0.034	-0.38	$+0.30$
LLaDA base / infill	$+0.041^*$	$+0.009^*$	-0.027^*	-0.074	$+0.11$	$+0.23$
LLaDA base / suffix	$+0.189^*$	$+0.063^*$	-0.042^*	-0.068	-0.04	$+0.08$
LLaDA instr. / infill	$+0.041^*$	$+0.010^*$	-0.017^*	-0.057	-0.05	$+0.06$
LLaDA instr. / suffix	$+0.144^*$	$+0.046^*$	-0.012	-0.072	-0.18	$+0.22$

On the random blocks, the per-order KL cost exceeds $\widehat{\mathrm{C}}$ by a median of 0.4 to 1.1 bits on prose and in the deployed regime of LLaDA-base and Dream-base (6 to 13 blocks per cell), $\widehat{\mathrm{C}}$ ranks blocks like it (Spearman 0.95 and 0.90, pooled across settings), the median order information is at most 0.98 bits, and on Dream-instruct prose the gap is 4.4 and 5.6 bits.

The offset gate also holds on code: on HumanEval windows it selects the documented convention by 12.6 to 15.9 bits, against 6.9 to 11.0 on prose. At a wrong offset greedy decoding of prose collapses on all three checkpoints, and the calibrated decode never repeats a token more than 3 times in a row. Scoring a block alone instead of in a batch of eight leaves LLaDA's entropies unchanged and moves Dream's by up to 3.3 bits at a single position. Gate 2 with its entropy tolerance (`scripts/verify/gate2.py`, 16 prose and code windows in both regimes) passes on all four checkpoints: the estimator's two batch shapes give identical entropies and argmaxes, while scoring a sequence alone moves Dream-base by up to 11.3 bits of entropy at one position. Taking B from the estimator's own first forward pass instead of the batch of eight changes 2 of the 197 impossibility flags, both on Dream-instruct, and 80 of the 101 blocks the main run flagged as outside $[0, W]$ are outside it again (39 of 53 outside Dream-instruct). On the near-deterministic blocks flagged below zero, the KL version, whose expectation cannot be negative, is below minus two standard errors on 16 of 38, by up to 0.23 bits. The per-block standard error under-covers there: when the sampled paths nearly coincide it is close to zero and misses the rare branches that keep the expectation nonnegative. B for identical blocks differs between the A100 and the GB10 by a mean of 0.04 to 0.19 bits per setting and by up to 3.5 bits on single Dream blocks.

Table 14: **Both terms of** (6)**, measured on a subset.** Medians over probed blocks, in bits, on a second GPU (NVIDIA GB10, same checkpoints and blocks as the main run). Per model and regime the probe takes 24 blocks the main run flagged as outside $[0, W]$, where there are that many, and 16 random others. $\widehat{\mathrm{C}}$ is the order-averaged estimate with B from the estimator's own first forward pass. *KL* is $\mathbb{E}_\pi \mathrm{KL}(q_\pi \parallel \prod_i p_i)$, estimated from the same samples by scoring them under the parallel marginals. I is the order information $I(\pi; Y_S)$, from all 4! orders evaluated at 32 sampled assignments per block. *KL*$-I$ is the cost relative to a random-order sampler. Where the conditionals compose the four columns agree. The last three columns use the random blocks only (cells with at least 3), which the text quotes: their number, the median of KL$-\widehat{\mathrm{C}}$, and the median I.

	all probed blocks					random blocks only		
Model / regime / domain	n	$\widehat{\mathrm{C}}$	KL	$I(\pi; Y_S)$	KL$-I$	n	KL$-\widehat{\mathrm{C}}$	I
Dream base / infill / prose	12	+0.55	+1.24	0.06	+1.03	10	+0.52	0.06
Dream base / infill / code	28	+0.00	+0.11	0.01	−0.02	6	−0.04	0.00
Dream base / suffix / prose	9	+7.60	+9.14	0.98	+7.75	9	+1.01	0.98
Dream base / suffix / code	11	+1.91	+1.35	0.23	+1.13	7	+0.41	0.38
Dream instr. / infill / prose	18	+0.14	+5.87	0.88	+5.05	12	+4.45	0.97
Dream instr. / infill / code	22	−0.00	+0.01	0.01	−0.00	4	+1.47	0.16
Dream instr. / suffix / prose	11	−0.72	+8.44	1.10	+6.95	6	+5.56	0.84
Dream instr. / suffix / code	29	+1.18	+1.42	0.14	+1.28	10	+0.18	0.16
LLaDA base / infill / prose	15	+1.99	+2.40	0.18	+2.06	13	+0.47	0.28
LLaDA base / infill / code	25	−0.00	−0.00	0.00	−0.02	3	−0.31	0.01
LLaDA base / suffix / prose	11	+6.77	+7.91	0.37	+7.77	10	+1.09	0.39
LLaDA base / suffix / code	6	+3.67	+3.88	0.15	+3.96	6	+1.15	0.15